Lecture Five

1 Exponential and Logarithmic Functions

1.1 Exponential Function

Definition 1.1.1 *Let $a > 0,\ a \neq 1$, then the exponential function of the base a ia $y = a^x$.*

1.1.1 Some Properties of the exponential function

For $y = a^x$,

1. If $a > 1$, then:
   (a) $a^x > 0\ ,\forall x \in \mathbb{R}$
   (b) $y = a^x$ is an increasing function.
   (c) $\lim\limits_{x\to-\infty} a^x \to 0$
   (d) $\lim\limits_{x\to+\infty} a^x \to \infty$
   (e) $f(0) = a^0 = 1,\ \forall a > 1$, the graph of a^x passes through the point (0,1) on $y-$axis.
2. If $0 < a < 1$, then:
   (a) $a^x > 0\ ,\forall x \in \mathbb{R}$
   (b) $y = a^x$ will be decreasing function.
   (c) $\lim\limits_{x\to-\infty} a^x \to \infty$
   (d) $\lim\limits_{x\to+\infty} a^x \to 0$
   (e) $f(0) = a^0 = 1,\ \forall a < 1$, the graph of a^x passes through the point (0,1) on $y-$axis.

Theorem 1.1.1 *Prove that*

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^k}{k!} + \cdots = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Proof

We know $e^x = \lim\limits_{n\to\infty}\left(1 + \frac{x}{n}\right)^n$, *now:*

$$\begin{aligned}\left(1+\frac{x}{n}\right)^n &= 1 + \binom{n}{1}\frac{x}{n} + \binom{n}{2}\frac{x^2}{n^2} + \binom{n}{3}\frac{x^3}{n^3} + \cdots + \binom{n}{k}\frac{x^k}{n^k} + \ldots \\ &= 1 + n.\frac{x}{n} + \frac{n(n-1)}{2!}\frac{x^2}{n^2} + \frac{n(n-1)(n-2)}{3!}\frac{x^3}{n^3} + \cdots + \frac{n(n-1)(n-2)\ldots}{k!}\frac{x^k}{n^k} + \ldots \\ &=1 + x + \frac{1}{2!}\left(1-\frac{1}{n}\right)x^2 + \frac{1}{3!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)x^3 + \cdots + \\ &\quad \frac{1}{k!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\ldots\left(1-\frac{k-1}{n}\right)x^k\end{aligned}$$

Therefore $\left(1+\dfrac{x}{n}\right)^n = \displaystyle\sum_{k=0}^{\infty} \frac{x^k}{k!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\dots\left(1-\frac{k-1}{n}\right)$ *and this implies that:*

$$\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = \sum_{k=0}^{\infty}\lim_{n\to\infty}\frac{x^k}{k!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\dots\left(1-\frac{k-1}{n}\right) =$$

$$\boxed{\Rightarrow e^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = \sum_{k=0}^{\infty}\frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots}$$

Example 1.1.1 *Evaluate the following sums using the definition of exponential function e^x:*

1. $\displaystyle\sum_{k=1}^{\infty}\frac{2^k}{k!}$

Sol

$$\sum_{k=1}^{\infty}\frac{2^k}{k!} = \left(1+\sum_{k=1}^{\infty}\frac{2^k}{k!}\right) - 1 = \frac{2^0}{0!} + \sum_{k=1}^{\infty}\frac{2^k}{k!} - 1 = \sum_{k=0}^{\infty}\frac{2^k}{k!} - 1 = e^2 - 1.$$

2. $\displaystyle\sum_{k=2}^{\infty}\frac{(-1)^k}{k!}$

Sol

$$\sum_{k=2}^{\infty}\frac{(-1)^k}{k!} = 1 - 1 + \sum_{k=2}^{\infty}\frac{(-1)^k}{k!} = \frac{(-1)^0}{0!} + \frac{(-1)^1}{1!} + \sum_{k=2}^{\infty}\frac{(-1)^k}{k!} = \sum_{k=0}^{\infty}\frac{(-1)^k}{k!} = e^{-1}.$$

3. $\displaystyle\sum_{k=2}^{\infty}\frac{(-1)^k}{2^{k-1}k!}$

Sol

$$\sum_{k=2}^{\infty}\frac{(-1)^k}{2^{k-1}k!} = \sum_{k=2}^{\infty}2\times\frac{(-1)^k}{4^k k!} = 2\times\sum_{k=2}^{\infty}\frac{\left(\frac{-1}{4}\right)^k}{k!} = 2\times\left[\sum_{k=2}^{\infty}\frac{\left(\frac{-1}{4}\right)^k}{k!} - 1 + \frac{1}{4}\right]$$

$$= 2\left(e^{-\frac{1}{4}} - \frac{3}{4}\right) = \frac{2}{\sqrt[4]{e}} - \frac{3}{2}.$$

Example 1.1.2 *Find the following limits using e^x definition.*

1. $\displaystyle\lim_{x\to 0}\frac{1-e^{2x}+2x}{x^2}$

Sol

We know that $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

This implies that $e^{2x} = 1 + 2x + \frac{4x^2}{2!} + \frac{8x^3}{3!} + \dots = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \dots$. *Now:*

$$\frac{1 - e^{2x} + 2x}{x^2} = \frac{1 + 2x - \left(1 + 2x + 2x^2 + \frac{4}{3}x^3 + \dots\right)}{x^2} = -\frac{\left(2x^2 + \frac{4}{3}x^3 + \dots\right)}{x^2}$$

$$= -\left(2 + \frac{4}{3}x + \dots\right)$$

$$\therefore \lim_{x\to 0} \frac{1 - e^{2x} + 2x}{x^2} = -\lim_{x\to 0}\left(2 + \frac{4}{3}x + \dots\right) = -2.$$

2. $\lim_{x\to 0} \frac{e^{ax} - 1}{\sin bx} \quad , b \neq 0$

Sol

We can write the numerator as follows:

$$e^{ax} - 1 = \sum_{k=0}^{\infty} \frac{(ax)^k}{k!} - 1 = \left(1 + ax + \frac{a^2x^2}{2!} + \frac{a^3x^3}{3!} + \dots\right) - 1 = ax + \frac{a^2x^2}{2!} + \frac{a^3x^3}{3!} + \dots$$

$$\therefore \lim_{x\to 0} \frac{e^{ax} - 1}{\sin bx} = \lim_{x\to 0} \frac{ax + \frac{a^2x^2}{2!} + \frac{a^3x^3}{3!} + \dots}{\sin bx} = \lim_{x\to 0} \frac{\frac{ax + \frac{a^2x^2}{2!} + \frac{a^3x^3}{3!} + \dots}{x}}{\frac{\sin bx}{x}}$$

$$= \frac{\lim_{x\to 0} \frac{ax + \frac{a^2x^2}{2!} + \frac{a^3x^3}{3!} + \dots}{x}}{\lim_{x\to 0} \frac{\sin bx}{x}} = \frac{a}{b}$$

3. $\lim_{x\to 0} \frac{a^x - b^x}{x}$

Sol

For you

1.1.2 Derivative of the Exponential Function

1.
$$\boxed{\frac{d}{dx}\left[a^{u(x)}\right] = a^{u(x)} \times \ln a \times \frac{du}{dx}}$$

2.
$$\boxed{\frac{d}{dx}\left[e^{u(x)}\right] = e^{u(x)} \times \ln e \times \frac{du}{dx} = u^{'}(x)e^{u(x)}}$$

And when $u(x) = x \Rightarrow$

1.
$$\frac{d}{dx}[a^x] = a^x \ln a$$

2.
$$\frac{d}{dx}[e^x] = e^x$$

1.2 Logarithmic Functions

Definition 1.2.2 *Let $a > 0$, $a \neq 1$. The* ***logarithmic function*** *of the base a is $y = \log_a x$, and it is the inverse of $y = a^x$. The domain of definition of it $D = \mathbb{R}^+ = (0, \infty)$ and $R = \mathbb{R}$.*

Note that if the base is e, it is called the natural logarithm, and we denoted by $\ln x$ ($\log_e x = \ln x$) and thus $\ln x$ is the inverse of e^x.

1.2.1 Properties:

No.	**log**	**ln**
(1)	$\log_a 1 = 0$	$\ln 1 = 0$
(2)	$\log_a x + \log_a y = \log_a xy$	$\ln x + \ln y = \ln xy$
(3)	$\log_a x^n = n \log_a x$	$\ln x^n = n \ln x$
(4)	$\log_a \frac{1}{x} = -\log_a x$	$\ln \frac{1}{x} = -\ln x$
(5)	$\log_a x - \log_a y = \log_a \frac{x}{y}$	$\ln x - \ln y = \ln \frac{x}{y}$
(6)	$\log_a b = \frac{1}{\log_b a}$	$\ln b = \frac{1}{\log_b e}$
(7)	$\log_{a^n} x = \frac{1}{n} \log_a x$	$\log_{e^n} x = \frac{1}{n} \ln x$
(8)	$\log_{a^n} x^n = \log_a x$	$\log_{e^n} x^n = \ln x$

1.2.2 Derivative of Logarithmic Function

1.
$$\boxed{\frac{d}{dx}[\log_a u(x)] = \frac{1}{u(x) \ln a} \times \frac{du}{dx}}$$

2.
$$\boxed{\frac{d}{dx}[\ln u(x)] = \frac{1}{u(x)} \times \frac{du}{dx}}$$

And if $u(x) = x \Rightarrow$

1.
$$\boxed{\frac{d}{dx}[\log_a x] = \frac{1}{x \ln a}}$$

2.
$$\boxed{\frac{d}{dx}[\ln x] = \frac{1}{x}}$$

Derivative of $[f(x)]^{g(x)}$:
Let $y = (f(x))^{g(x)}$, then the derivative evaluated as follows:

$$y = [f(x)]^{g(x)} \Rightarrow \ln y = g(x) \ln f(x)$$

Now, by differentiating both sides w.r.t x, we get:

$$\frac{y'}{y} = g(x) \times \frac{f'(x)}{f(x)} + g'(x) \ln f(x)$$

Therefore the derivative will be:

$$\boxed{y' = \left[g(x) \times \frac{f'(x)}{f(x)} + g'(x) \ln f(x) \right] \times y}$$

Example 1.2.3 *Find the first derivative of the following function:*

1. $y = e^{\sin^2(4x)}$

Sol

$$\frac{dy}{dx} = 2(\sin 4x) \times (4 \cos 4x) \times e^{\sin^2(4x)} = 8 \sin 4x \cos 4x \; e^{\sin^2(4x)}.$$

2. $y = e^{\frac{2x^2+8}{4x+1}}$

Sol

$$\frac{dy}{dx} = \left(\frac{(4x+1)(4x) - (2x^2+8)(4)}{(4x+1)^2} \right) e^{\frac{2x^2+8}{4x+1}} = \frac{4x - 8x^2 - 12}{(4x+1)^2} e^{\frac{2x^2+8}{4x+1}}.$$

3. $y = x^3 e^{2x^2-1}$

Sol

$$\frac{dy}{dx} = x^3 \left(4x \times e^{2x^2-1} \right) + e^{2x^2-1} \times (3x^2) = 4x^4 e^{2x^2-1} + 3x^2 e^{2x^2-1}$$

4. $y = e^{\frac{\tan x}{\sin(\cos x)}}$

Sol

$$\frac{dy}{dx} = \left[\frac{\sin(\cos x) \sec^2 x + \tan x (\sin x \cos(\cos x))}{\sin^2(\cos x)} \right] e^{\frac{\tan x}{\sin(\cos x)}}$$

Example 1.2.4 *Find* $\frac{dy}{dx}$ *for the following functions:*

(a) $y = 2^x + \frac{1}{\sqrt{3^x}}$

Sol

$$y = 2^x + 3^{-\frac{x}{2}} \Rightarrow \frac{dy}{dx} = 2^x \ln 2 + 3^{-\frac{x}{2}} \times \ln 3 \times \left(-\frac{1}{2} \right) = 2^x \ln 2 - \frac{\ln 3}{2\sqrt{3^x}}$$

(b) $y = \frac{4^x - 1}{4^x + 1}$

Sol

$$\frac{dy}{dx} = \frac{(4^x+1)(4^x\ln 4) - (4^x-1)(4^x\ln 4)}{(4^x+1)^2}$$
$$= \frac{4^x\ln 4(4^x+1-4^x+1)}{(4^x+1)^2} = \frac{2^{2x+1}\ln 4}{(4^x+1)^2}$$

(c) $y = x^{\sec^{-1}x}$

Sol

By tanking $\ln$ *for both sides* $\Rightarrow$ $\ln y = \sec^{-1}x\ln x$, *now:*

$$\frac{1}{y}\frac{dy}{dx} = \frac{sec^{-1}x}{x} + \frac{\ln x}{|x|\sqrt{x^2-1}}$$

Therefore, the required derivative is:

$$\frac{dy}{dx} = \left[\frac{sec^{-1}x}{x} + \frac{\ln x}{|x|\sqrt{x^2-1}}\right]x^{\sec^{-1}x}$$

(d) $y = \dfrac{(x+1)^3(x^2+4)^8}{\sqrt[4]{x^2+2}}$

Sol

Firstly we take the $\ln$ *for both sides:*

$$\ln y = \ln\left[\frac{(x+1)^3(x^2+4)^8}{\sqrt[4]{x^2+2}}\right]$$
$$= 3\ln(x+1) + 8\ln(x^2+4) - \frac{1}{4}\ln(x^2+2)$$

Then we differentiate both sides, we get:

$$\frac{1}{y}\frac{dy}{dx} = \frac{3}{x+1} + 8\times\frac{2x}{x^2+4} - \frac{1}{4}\frac{2x}{x^2+2}$$

Finally, we multiply both sides by y *to obtain the derivative:*

$$\frac{dy}{dx} = \frac{(x+1)^3(x^2+4)^8}{\sqrt[4]{x^2+2}}\left[\frac{3}{x+1} + \frac{16x}{x^2+4} - \frac{x}{2(x^2+2)}\right]$$

(e) $x^{\sin y} = [\sin(x+y)]^y$

Sol

By taking the $\ln$ *for both sides, the given function can be written as:*

$$\sin y\ln x = y\ln(\sin(x+y))$$

Now, we differentiate both sides:

$$\frac{\sin y}{x} + y^{'}\cos(y)\ln x = y\frac{\cos(x+y)}{\sin(x+y)}\left(1+y^{'}\right)$$

Therefore:

$$y^{'} = \frac{y\cot(x+y) - \dfrac{\sin y}{x}}{\cos(y)\ln x - \cot(x+y)}$$

(f) $x = 2^y - 2^{-y}$

Sol

Differentiate both sides directly:

$$1 = 2^y \ln 2 \, \frac{dy}{dx} + 2^{-y} \ln 2 \, \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{(2^y + 2^{-y}) \ln 2}$$

Exercise 1.2.1 *1. Find* $\frac{dy}{dx}$ *for the following:*

(i) $y = \pi^x - x^\pi$

Sol

(ii) $y = e^{\sin 3x}$

Sol

(iii) $y = \ln(x^2 - 1)$

Sol

(iv) $y = \ln\left(\frac{2^x - x}{2^x + x}\right)$

Sol

(v) $y = \log_3\left(\frac{x}{2^x - x}\right)$

Sol

(vi) $y = \tan^{-1}\left(e^{-x^2}\right)$

Sol

(vii) $y = \sin\sqrt{e^x - 1}$

Sol

(viii) $y = \dfrac{\ln(x) + 1}{\ln(x) - 1}$

Sol

(ix) $y = \dfrac{(x+1)^3\sqrt{x^2+4}}{\sqrt[3]{x+2}}$

Sol

(x) $y = \ln\left[\ln\left(\dfrac{x+1}{x-1}\right)\right]$

Sol

(xi) $y = \dfrac{\sin^4(x)\cos^3(x)}{\sqrt{x}}$

Sol

(xii) $xy = 2^{x-y}$

Sol

(xiii) $y + x = y^x$

Sol

2. *Find the following limits:*

   (a) $\lim\limits_{x \to 1} \dfrac{\ln x}{x-1}$

   Sol

   (b) $\lim\limits_{x \to 0} \dfrac{1-e^{2x}}{x}$

   Sol

   (c) $\lim\limits_{x \to 0} \dfrac{e^{ax}-e^{bx}}{x}$

   Sol

   (d) $\lim\limits_{x \to 0} \dfrac{e^{-x}-e^{2x}}{x}$

   Sol

   (e) $\lim\limits_{x \to 0} \dfrac{e^{-x^2}-1}{x}$

   Sol